Targeted latent tuberculosis infection screening in the APY Land

Working health-economic analysis for SA Health planning and refinement

Analysis date: 02 September 2026

Commit: 12f697aa7caccad9a9290a7c7eef2dad54257e46

This report is provisional and supports planning. It does not claim cost-effectiveness.

Executive summary

Measure: Eligible population; Value: 1,500; Interpretation: people

Measure: People screened; Value: 450; Interpretation: people

Measure: Active TB cases averted; Value: 12; Interpretation: cases over 20 years

Measure: Included gross delivery expenditure; Value: AUD 95,052; Interpretation: AUD 2019, 3% discounted

Measure: Active-TB care cost offset; Value: AUD 187,422; Interpretation: AUD 2019, 3% discounted

Measure: Net included health-system result; Value: AUD 92,370 net saving; Interpretation: AUD 2019, 3% discounted

Measure: Provisional DALYs averted; Value: 16.2; Interpretation: secondary, provisional

Measure: Willingness-to-pay threshold; Value: Not supplied; Interpretation: NMB not calculated

Reference strategy and methods

Comparator: current practice, including ordinary background TB diagnosis and care, without an additional systematic LTBI screening programme.

Intervention: targeted systematic LTBI screening with IGRA and 3HP preventive treatment, prioritising people most likely to avoid active TB.

Population: all 1500 APY demonstration residents eligible; 30.0% screened over 2 years with 20 years of follow-up.

Economic perspective: Australian health-care system; currency and price year: AUD 2019; primary discounting: 3% for costs and DALYs, with 0% retained for comparison.

The primary epidemiological anchor is a stochastic individual-level compatibility run reproducing the frozen earlier APY reference analysis.

The model simulates the population, prioritises screening using the active-TB-prevention targeting rule, applies test performance, treatment initiation, completion, adverse-event stopping and treatment efficacy, and records events in a shared ledger.

Costs are calculated from event-ledger quantities: screening tests per person screened, pathway costs per screened person or treatment start, regimen costs per treatment start, ADR management per ADR-related stop, and active-TB care per active-TB case.

Annual costs and DALYs are discounted by model year. Quantities such as people screened and TB cases are not discounted.

DALYs use acute active-TB disability and mortality assumptions. Treatment, ADR and post-TB health losses are excluded from the primary analysis through explicit reviewed-exclusion records.

Simulation percentiles describe finite-population variation under fixed parameter assumptions. They are not confidence intervals, credible intervals or a probabilistic sensitivity analysis.

Economic inputs

Cost item: IGRA screening test per person; Value: AUD 113; Unit: cost per person screened; Status: Reviewed working numerical assumption; Caveat: Supports current cost-consequence calculation.

Cost item: 3HP preventive regimen per started course; Value: AUD 166; Unit: cost per started course; Status: Reviewed working numerical assumption; Caveat: Supports current cost-consequence calculation.

Cost item: Active-TB disease care per case; Value: AUD 19,080; Unit: cost per active TB case; Status: Reviewed working numerical assumption; Caveat: Supports current cost-consequence calculation.

Cost item: TPT ADR management per ADR stop; Value: AUD 39; Unit: cost per ADR stop; Status: Reviewed working numerical assumption; Caveat: Supports current cost-consequence calculation.

Cost item: Return/results pathway cost per screened person; Value: AUD 50; Unit: AUD per person screened; Status: Reviewed working numerical assumption; Caveat: Supports current cost-consequence calculation.

Cost item: Clinical-review pathway cost per TPT start; Value: AUD 50; Unit: AUD per TPT start; Status: Reviewed working numerical assumption; Caveat: Supports current cost-consequence calculation.

Cost item: Active-TB exclusion / CXR / laboratory work-up per TPT start; Value: AUD 150; Unit: AUD per TPT start; Status: Reviewed working numerical assumption; Caveat: Supports current cost-consequence calculation.

Cost item: False-positive incremental care per false-positive TPT start; Value: Reviewed as bundled/no additional cost; Unit: cost per false-positive TPT started; Status: Reviewed as bundled/no additional cost; Caveat: Supports current cost-consequence calculation.

Cost item: Programme setup total cost; Value: Reviewed exclusion; Unit: total once at programme start; Status: Reviewed exclusion; Caveat: Not yet locally costed; zero compatibility value in the primary working reference.

Cost item: Programme running cost; Value: Reviewed exclusion; Unit: annual or total over screening programme; Status: Reviewed exclusion; Caveat: Not yet locally costed; zero compatibility value in the primary working reference.

Cost item: Travel, outreach and staff-support cost per screened person; Value: AUD 0; Unit: AUD per person screened; Status: Reviewed working numerical assumption; Caveat: Not yet locally costed; zero compatibility value in the primary working reference.

Epidemiological results

Measure: Eligible population; Value: 1,500; Interpretation: people

Measure: Screened; Value: 450; Interpretation: people

Measure: True-positive latent results; Value: 65.9; Interpretation: people

Measure: False-positive results; Value: 7.4; Interpretation: people

Measure: Preventive treatments initiated; Value: 62.2; Interpretation: people

Measure: Preventive treatments completed; Value: 49.7; Interpretation: people

Measure: ADR-related treatment stops; Value: 3.1; Interpretation: people

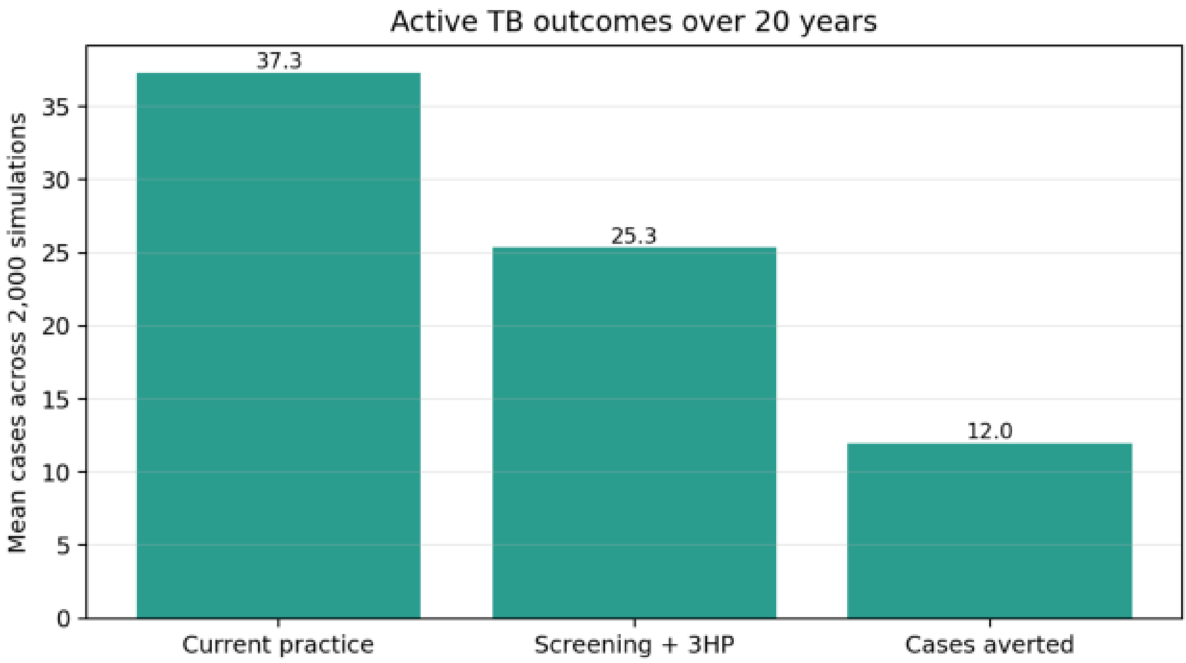
Measure: Infections effectively treated; Value: 36.7; Interpretation: people

Measure: Comparator active TB; Value: 37.3; Interpretation: cases

Measure: Intervention active TB; Value: 25.3; Interpretation: cases

Measure: Active TB cases averted; Value: 12.0; Interpretation: cases

Measure: Relative reduction; Value: 32.1%; Interpretation: direct active-TB outcomes



Cost categories

Category: Screening test; Comparator: AUD 0; Intervention: AUD 50,323; Incremental: AUD 50,323; Interpretation: Included mapped event cost.

Category: Return/results; Comparator: AUD 0; Intervention: AUD 22,173; Incremental: AUD 22,173; Interpretation: SA Health pathway working cost; possible bundle overlap should be reviewed.

Category: Clinical review; Comparator: AUD 0; Intervention: AUD 3,069; Incremental: AUD 3,069; Interpretation: SA Health pathway working cost; possible bundle overlap should be reviewed.

Category: Active-TB exclusion/work-up; Comparator: AUD 0; Intervention: AUD 9,208; Incremental: AUD 9,208; Interpretation: SA Health pathway working cost; possible bundle overlap should be reviewed.

Category: Preventive regimen; Comparator: AUD 0; Intervention: AUD 10,159; Incremental: AUD 10,159; Interpretation: Included mapped event cost.

Category: ADR management; Comparator: AUD 0; Intervention: AUD 120; Incremental: AUD 120; Interpretation: Included mapped event cost.

Category: Programme setup; Comparator: AUD 0; Intervention: AUD 0; Incremental: AUD 0; Interpretation: Not yet locally costed in primary reference.

Category: Programme running; Comparator: AUD 0; Intervention: AUD 0; Incremental: AUD 0; Interpretation: Not yet locally costed in primary reference.

Category: Travel/outreach/staff; Comparator: AUD 0; Intervention: AUD 0; Incremental: AUD 0; Interpretation: Not yet locally costed in primary reference.

Category: Active-TB care; Comparator: AUD 626,261; Intervention: AUD 438,839; Incremental: AUD -187,422; Interpretation: Offset from fewer modelled active-TB cases.

Category: Total; Comparator: AUD 626,261; Intervention: AUD 533,891; Incremental: AUD -92,370; Interpretation: Included mapped event cost.

Gross and net ratios

Measure: Intervention delivery expenditure per person screened; Numerator: AUD 95,052; Denominator: 450.0; Result: AUD 211

Measure: Intervention delivery expenditure per TPT start; Numerator: AUD 95,052; Denominator: 62.2; Result: AUD 1,527

Measure: Intervention delivery expenditure per TPT completion; Numerator: AUD 95,052; Denominator: 49.7; Result: AUD 1,912

Measure: Intervention delivery expenditure per infection effectively treated; Numerator: AUD 95,052; Denominator: 36.7; Result: AUD 2,589

Measure: Intervention delivery expenditure per active TB case averted; Numerator: AUD 95,052; Denominator: 12.0; Result: AUD 7,943

Measure: Net health-system saving per person screened; Numerator: AUD 92,370 net saving; Denominator: 450.0; Result: AUD 205 saving

Measure: Net health-system saving per TPT completion; Numerator: AUD 92,370 net saving; Denominator: 49.7; Result: AUD 1,858 saving

Measure: Net health-system saving per active TB case averted; Numerator: AUD 92,370 net saving; Denominator: 12.0; Result: AUD 7,719 saving

Budget impact

Year: 0; Delivery expenditure: AUD 48,938; Comparator TB care: AUD 220,675; Intervention TB care: AUD 206,727; Annual incremental: AUD 34,990; Cumulative incremental: AUD 34,990

Year: 1; Delivery expenditure: AUD 46,111; Comparator TB care: AUD 146,691; Intervention TB care: AUD 101,140; Annual incremental: AUD 560; Cumulative incremental: AUD 35,551

Year: 2; Delivery expenditure: AUD 4; Comparator TB care: AUD 23,946; Intervention TB care: AUD 12,076; Annual incremental: AUD -11,866; Cumulative incremental: AUD 23,685

Year: 3; Delivery expenditure: AUD 0; Comparator TB care: AUD 22,402; Intervention TB care: AUD 11,105; Annual incremental: AUD -11,297; Cumulative incremental: AUD 12,388

Year: 4; Delivery expenditure: AUD 0; Comparator TB care: AUD 21,292; Intervention TB care: AUD 10,417; Annual incremental: AUD -10,875; Cumulative incremental: AUD 1,513

Year: 5; Delivery expenditure: AUD 0; Comparator TB care: AUD 19,075; Intervention TB care: AUD 9,398; Annual incremental: AUD -9,677; Cumulative incremental: AUD -8,164

Year: 6; Delivery expenditure: AUD 0; Comparator TB care: AUD 17,888; Intervention TB care: AUD 8,868; Annual incremental: AUD -9,020; Cumulative incremental: AUD -17,184

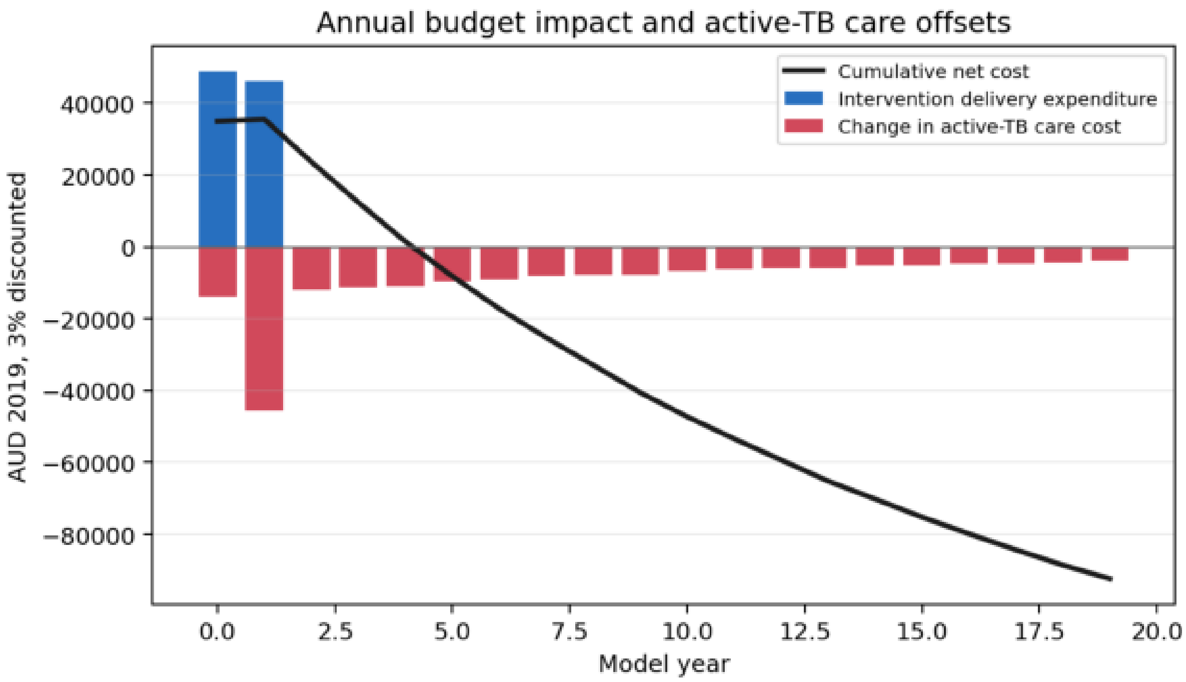
Year: 7; Delivery expenditure: AUD 0; Comparator TB care: AUD 16,289; Intervention TB care: AUD 8,214; Annual incremental: AUD -8,075; Cumulative incremental: AUD -25,259

Year: 8; Delivery expenditure: AUD 0; Comparator TB care: AUD 15,416; Intervention TB care: AUD 7,779; Annual incremental: AUD -7,636; Cumulative incremental: AUD -32,895

Year: 9; Delivery expenditure: AUD 0; Comparator TB care: AUD 15,062; Intervention TB care: AUD 7,421; Annual incremental: AUD -7,640; Cumulative incremental: AUD -40,536

Year: 10; Delivery expenditure: AUD 0; Comparator TB care: AUD 13,799; Intervention TB care: AUD 7,077; Annual incremental: AUD -6,722; Cumulative incremental: AUD -47,258

Year: 11; Delivery expenditure: AUD 0; Comparator TB care: AUD 12,695; Intervention TB care: AUD 6,561; Annual incremental: AUD -6,134; Cumulative incremental: AUD -53,392



Scenarios and break-even analysis

Scenario: Primary working reference; Incremental cost: AUD 92,370 net saving; Active TB cases averted: 11.97; DALYs averted: 16.2; Interpretation: Primary working reference; local programme resources remain not locally costed.

Scenario: Illustrative AUD 500,000 setup; Incremental cost: AUD 407,630 net cost; Active TB cases averted: 11.97; DALYs averted: 16.2; Interpretation: Adds illustrative setup cost; epidemiological outcomes unchanged.

Scenario: Bundled/excluded pathway costs; Incremental cost: AUD 126,819 net saving; Active TB cases averted: 11.97; DALYs averted: 16.2; Interpretation: Excludes selected pathway components to test possible overlap with bundled costs.

Scenario: Higher post-TB DALY burden; Incremental cost: AUD 92,370 net saving; Active TB cases averted: 11.97; DALYs averted: 73.1; Interpretation: Costs unchanged; DALY gain increases under secondary post-TB burden assumption.

Question: One-off additional setup cost before net saving is exhausted; Value: AUD 92,370; Interpretation: Arithmetic break-even only; not a willingness-to-pay threshold.

Question: Equivalent annual running cost over two screening years; Value: AUD 46,867; Interpretation: Assumes equal payments in years 0 and 1 with 3% cost discounting.

Limitations and next information required

The inherited 10/770 active-TB calibration target has unresolved provenance and interpretation.

The implicit early phase is retained for compatibility with the earlier analysis, not interpreted as measured recent LTBI.

Disease-risk odds ratios are used as multiplicative hazard multipliers for compatibility and remain scientifically provisional.

Active or near-baseline TB is not fully separated from future incident, preventable TB in the compatibility model.

Programme setup, programme running, travel, outreach and staffing costs are not yet locally costed.

Some pathway costs may overlap with bundled test or treatment costs and require review.

DALY assumptions are working assumptions and the ICER result is secondary and provisional.

No reviewed willingness-to-pay threshold is supplied, so NMB and probability cost-effective are unavailable.

Transmission and indirect community effects are excluded.

The compatibility model expects targeted IGRA/3HP screening to reduce direct active-TB outcomes in the simulated APY cohort.

Included delivery costs are offset by avoided active-TB care under the working assumptions, but this depends materially on unresolved local programme resources and inherited epidemiological calibration semantics.

The immediate next step is for SA Health to enter local setup, running, travel, outreach and workforce costs in the Streamlit Health Economics workspace and review possible bundling with test and regimen costs.

Validation and reproducibility

Metric: nScreened; Compatibility mean: 450.00; Compatibility median: 450.0; Frozen median: 450.0; Frozen 2.5-97.5%: 450.0 to 450.0

Metric: nTestPositive; Compatibility mean: 83.29; Compatibility median: 83.0; Frozen median: 84.0; Frozen 2.5-97.5%: 68.0 to 100.0

Metric: nTestPositiveNonActive; Compatibility mean: 73.28; Compatibility median: 73.0; Frozen median: 73.0; Frozen 2.5-97.5%: 59.0 to 89.0

Metric: nFalsePositiveTests; Compatibility mean: 7.40; Compatibility median: 7.0; Frozen median: 7.0; Frozen 2.5-97.5%: 3.0 to 13.0

Metric: nStartTPT; Compatibility mean: 62.24; Compatibility median: 62.0; Frozen median: 62.0; Frozen 2.5-97.5%: 48.0 to 77.0

Metric: nCompleteTPT; Compatibility mean: 49.72; Compatibility median: 50.0; Frozen median: 50.0; Frozen 2.5-97.5%: 37.0 to 63.0

Field: Repository; Value: github.com/emmamcbryde/tb-community-risk

Field: Branch; Value: codex/apy-sa-health-matlab-compatible-anchor

Field: Commit; Value: 12f697aa7caccad9a9290a7c7eef2dad54257e46

Field: Model type; Value: Stochastic individual-based compatibility anchor

Field: Simulations; Value: 2000

Field: Seed; Value: 1

Field: Population; Value: 1500

Field: Coverage; Value: 30.0%

Field: Screening window; Value: 2 years

Field: Follow-up; Value: 20 years